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RESEARCH ARTICLE

Molecular insights into expression and silencing of resistance determinants in *Staphylococcus aureus*

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Abstract

Objective: This study aimed to investigate the status of antimicrobial-resistant strains of *Staphylococcus aureus* in Pakistan, their association in terms of co-occurrence with the biofilm-forming genes, resistance profiling and associated discrepancies in diagnostic methods.

Methodology: A total of 384 milk samples from bovine was collected by using convenient sampling technique and were initially screened for subclinical mastitis, further preceded by isolation and confirmation of *S. aureus*. The *S. aureus* isolates were subjected to evaluation of antimicrobial resistance by phenotypic identification using Kirby–Bauer disc diffusion method, while the genotypic estimation was done by polymerase chain reaction to declare isolates as methicillin, beta-lactam, vancomycin, tetracycline, and aminoglycoside resistant *S. aureus* (MRSA, BRSA, VRSA, TRSA, and ARSA), respectively.

Results: The current study revealed an overall prevalence of subclinical mastitis and *S. aureus* to be 59.11% and 46.69%, respectively. On a phenotypic basis, the prevalence of MRSA, BRSA, VRSA, TRSA, and ARSA was found to be 44.33%, 58.49%, 20.75%, 35.84%, and 30.18%, respectively. The results of PCR analysis showed that 46.80% of the tested isolates were declared as MRSA, 37.09% as BRSA, and 36.36% as VRSA, while the occurrence of TRSA and ARSA was observed in 26.31% and 18.75%, respectively. The current study also reported the existence of biofilm-producing genes (*icaA* and *icaD*) in 49.06% and 40.57% isolates, respectively. Lastly, this study also reported a high incidence of discrepancies for both genotypic and phenotypic identification methods of resistance evaluation, with the highest discrepancy ratio for the *accA-aphD* gene, followed by *tetK*, *vanB*, *blaZ*, and *mecA* genes.

Conclusion: The study concluded that different antibiotic resistance strains of *S. aureus* are prevalent in study districts with high potential to transmit between human populations. The study also determined that there are multiple resistance determinants and mechanisms that are responsible for the silencing and expression of antibiotic resistance genes.

KEYWORDS
antimicrobial resistance, biofilm, discrepancies, *S. aureus*

INTRODUCTION

Antimicrobial resistance (AMR) is one of the major global health threats to humanity in the 21st century due to the dissemination of antibiotic resistance genes among antimicrobial-resistant organisms [1]. AMR is defined as the ability of the microorganisms to resist the antimicrobial ability of antibiotics that would previously kill those microorganisms or at the least prevent their further growth [2, 3]. Consequently, traditional antibiotic therapies have become largely ineffective with the persistence and spread of AMR [4]. If the issue of AMR remains unaddressed, it can lead to 10 million deaths globally by 2050 and a world Gross domestic product setback of $3.4 trillion by the next decade,

Sustainable Development Goal: Good Health and Well-being according to the UN Environment Programme [5]. AMR can be acquired by either horizontal gene transfer or mutation within the nuclear material of the microorganism [6, 7].

Staphylococcus aureus (*S. aureus*) is a ubiquitous organism responsible for a variety of maladies in humans and animals, ranging from simple skin infections to life-threatening conditions of sepsis [8–10]. *S. aureus* is a major mastitogen of dairy animals, responsible for the majority of the subclinical infections responsible for serious economic losses and, through contamination of dairy products, a significant threat to consumers [11, 12]. *S. aureus'* virulence is influenced by its ability to secrete toxins, produce extracellular enzymes, and the presence of surface proteins responsible for the disruption of host cell membranes, breakdown of host molecules for survival and immune evasion, respectively [9].

S. aureus possesses the ability to escape the biocidal actions of antimicrobial agents owning it being a part of the ESKAPE (*Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa,* and *Enterobacter* spp.) group [13]. *S. aureus* has developed resistance to antibacterial agents through restriction of drug absorption, drug targets, receptor modification, drug enzyme inactivation and drug outflow pathways. In particular, the localisation of resistance-encoding genes on mobile genetic islands such as transposons, plasmids and altering genetic constituents favours the horizontal as well as vertical transfer of resistance between interacting bacterial species [7, 14]. *S. aureus* has acquired the ability to develop resistance against various antibiotics, and the incidence of multidrug-resistant *S. aureus* (MDRSA) has severe concerns for public health [15, 16]. *S. aureus* genomic material harbours resistance genes including *blaZ*, *mecA* and *mecC*, *vanA* and *vanB*, *tetK* and *tetM*, *acca-aphD* encoding resistance for the β-lactam, methicillin, vancomycin, tetracycline and aminoglycosides group of antibiotics, respectively [17]. It also harbours a set of genes responsible for biofilm production, *icaA* and *icaD* along with other genes including *rbf*, *mgrA*. Biofilm production enhances the attachment of the organism to cell surfaces and hinders the absorption of the antibiotics as it forms a physical covering over the surface of the organism, leading to chronic persistent infections. Biofilm formation protects the organism from the external environment by forming a stable internal environment [18].

The development of resistance to various antibiotics by *S. aureus* is influenced by a variety of resistance determinants. The discrepancies found in the phenotypic and genotypic methods for diagnosis of antimicrobial-resistant *S. aureus* raise the need for an in-depth study of various determinants influencing the expression or silencing of AMR in *S. aureus*. Keeping in view the importance of AMR in animals and public health hazards, huge economic losses, and transmission of *S. aureus* at the human-animal interface, this study was planned to investigate different antimicrobial-resistant strains of *S. aureus* from bovine milk of three districts of Punjab, Pakistan, along with the associated manifestation of biofilm-forming genes. The study also highlighted the evaluation of any discrepancies between the genotypic and phenotypic methods for the diagnosis of AMR genes.

MATERIALS AND METHODS

Raw milk sample collection and assessment of subclinical mastitis

From December 2022 to April 2023, a total of raw milk samples ($n = 384$) were aseptically collected from apparently healthy bovines in three districts of Punjab-Pakistan, including Faisalabad, Rawalpindi, and Muzaffargarh, using a convenient sampling technique (Figure 1). Samples were subjected to the California mastitis test (CMT) for screening of subclinical mastitis (SCM), and positive milk samples were transferred to the Medicine Research Laboratory, Department of Veterinary Medicine, University of Veterinary and Animal Sciences, Lahore, Pakistan, maintaining the cold chain protocol.

Isolation, identification and confirmation of *S. aureus*

The SCM-positive milk samples were swabbed on 5% blood agar. The positive colonies of bacteria showing haemolysis on blood agar were subjected to Gram staining to confirm the presence of purple colour cocci. The isolation of *S. aureus* was done on the selective media, mannitol salt agar. The isolates showing mannitol fermentation as well as the yellow golden colonies were declared to be *S. aureus*, which was further checked for different biochemical tests, including catalase and coagulase tests [19]. Genotypic confirmation of *S. aureus* was made by targeting the *nuc* gene utilising primers sequence and condition mentioned in Table 1 [20].

Phenotypic identification of antibiotic resistance

Phenotypic assessment of antibiotic sensitivity or resistivity for all the genotypic *S. aureus* isolates was carried out utilising the Kirby–Bauer disc diffusion method. Mueller Hinton agar plates were prepared and streaked with active *S. aureus* growth (0.5 McFarland). Using a disc dispenser, various antibiotics discs (Bioanalyse®) like cefoxitin (30 μg), penicillin-G (10 μg), tetracycline (30 μg), vancomycin (30 μg) and gentamicin (10 μg) discs were placed and incubation at 35–37°C was provided for 24 h. The zone of inhibition (ZoI) formed by antibiotics was measured and the readings were compared with the standard values provided by the Clinical and Laboratory Standard Institute [27]. Based on ZoI by cefoxitin, penicillin, vancomycin, tetracycline and gentamicin discs, isolates were categorised as methicillin-resistant *S. aureus* (MRSA) or methicillin-sensitive *S. aureus* (MSSA), β-lactam-resistant *S. aureus* (BRSA) or β-lactam-sensitive *S. aureus* (BSSA), vancomycin-resistant *S. aureus* (VRSA) or vancomycin-sensitive *S. aureus* (VSSA), tetracycline-resistant *S. aureus* (TRSA) or tetracycline-sensitive *S. aureus* (TSSA), and

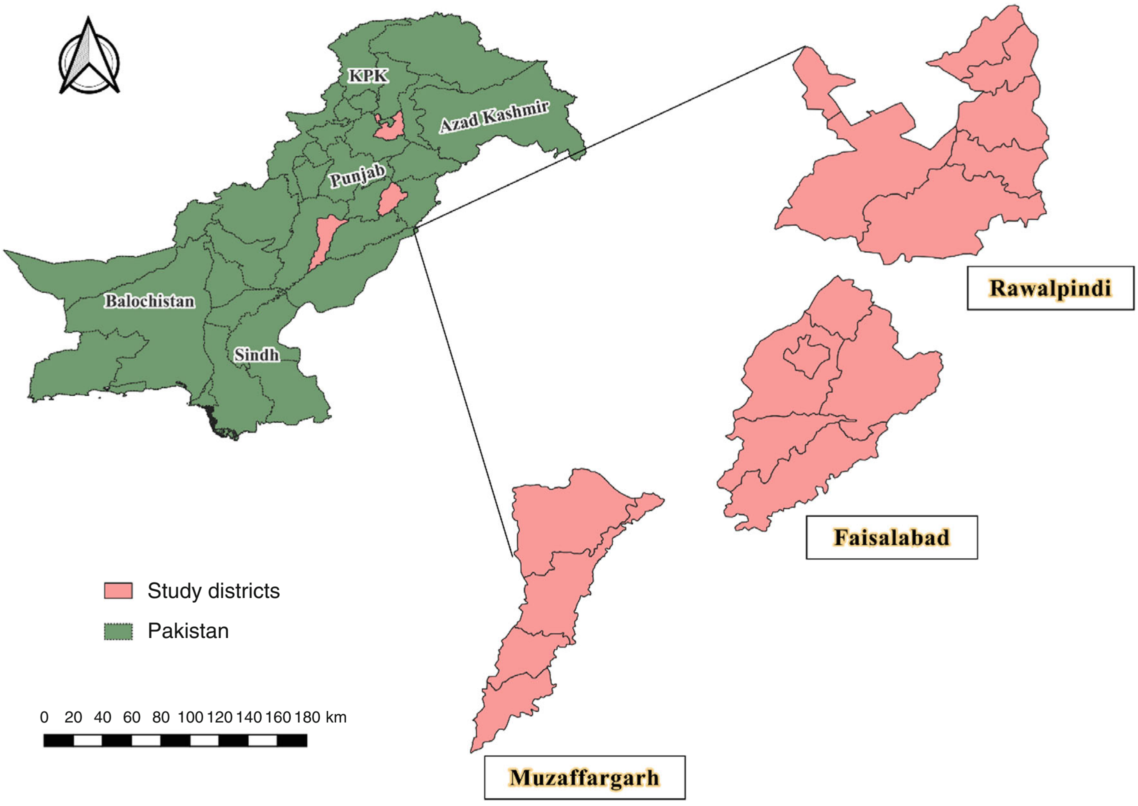

FIGURE 1 Quantum geographic information system (QGIS) map of the study districts.

aminoglycoside-resistant *S. aureus* (ARSA) or aminoglycoside-sensitive *S. aureus* (ASSA), respectively.

Genotypic assessment of antimicrobial-resistant genes

For molecular identification of resistance-encoding genes, a commercially available DNA Kit (Vivantis Technologies SDN, BHD, Malaysia) was used for the extraction of DNA from all the phenotypic sensitive and resistant *S. aureus* isolates [28]. Utilising nano-drop assay, quantification of the extracted DNA was also performed. Genotypic assessment was made by the presence of β-lactamase encoding gene (*blaZ*), methicillin resistance (*mecA*), vancomycin resistance (*vanB*), aminoglycoside acetyltransferase and aminoglycoside phosphotransferase (*aaacA-aphD*), and tetracycline efflux transporter (*tetK*) genes in *S. aureus* isolates through PCR to declare them as genotypic BRSA or BSSA, MRSA or MSSA, VRSA or VSSA, ARSA or ASSA, and TRSA or TSSA, respectively, utilising already reported primer sequences and conditions (Table 1). For each PCR reaction, a 20 μL PCR recipe was prepared, which included Master Mix (10 μL), forward and reverse primers (0.7 μL each), template DNA sample (2 μL) and diethyl pyrocarbonate water (up to a total volume of 20 μL). PCR amplified product was illuminated under a UV-illuminator after performing gel electrophoresis in a 1.5% agarose gel using ethidium bromide at 0.5 mg/mL as a dye and a 100 bp DNA ladder as the marker. The isolates showing the desired bands at specific base pairs for each particular gene (*blaZ* at 421 bp; *mecA* at 310 bp; *vanB* at 433 bp; *aaacA-aphd* at 228 bp and *tetK* at 697 bp) were considered positive for that gene, while the isolates that did not show any band at its required base pair length were declared negative for that gene.

Molecular identification of biofilm-producing genes

All the isolates after DNA extraction and quantification were amplified by PCR targeting intercellular adhesion genes (*icaA* and *icaD*), which are responsible for the formation of biofilm, using already reported primer sequences and conditions (Table 1).

Assessment of antibiotic resistance profile

To evaluate the phenotypic incidence of the resistance profile of *S. aureus* isolates, the Kirby–Bauer disc diffusion method was used. For this purpose, 45 isolates were randomly selected and tested against 13 commonly used antibiotics, levofloxacin (5 μg), linezolid (30 μg), tylosin (30 μg), gentamicin (10 μg), amikacin (30 μg), ciprofloxacin (5 μg), moxifloxacin (5 μg), amoxicillin (25 μg), oxytetracycline

TABLE 1 Primer sequence and annealing temperature.

			PCR conditions					
			Denaturation			Extension		
Target gene	Product size (bp)	Primer sequence ($5' \rightarrow 3'$)	Initial	Final	Annealing	Initial	Final	References
nuc	270	P1: GCGATTGATGGTGATACGGTT P2: AGCCAAGCCTTGACGAACTAAAGC	94°C for 3 min	94°C for 1.5 min	55°C for 2 min	-	72°C for 3 min	[20]
blaZ	421	P1: CAAAGATGATATAGTTGCTTATTCTCC P2: TGCTTGACCACTTTTATCAGC	94°C for 5 min	94°C for 30 s	52°C for 1 min	72°C for 1 min	72°C for 10 min	[21]
mecA	310	P1: TGGCATTCGTGTCACAATCG P2: CTGGAACTTGTTGAGCAGAG	94°C for 3 min	94°C for 1.5 min	58°C for 30 s	72°C for 30 s	72°C for 10 min	[22]
vanB	433	P1: GTGACAAACCGGAGGCGAGGA P2: CCGCCATCCTCCTGCAAAAAA	94°C for 5 min	94°C for 1 min	54°C for 1 min	72°C for 1 min	72°C for 10 min	[23]
tetK	697	P1: TTAGGTGAAGGGTTAGGTCC P2: GCAAACTCATTCCAGAAGCA	94°C for 3 min	94°C for 30 s	55°C for 30 s	72°C for 30 s	72°C for 4 min	[24]
accA-aphD	227	P1: TAATCCAAGAGCAATAAGGGC P2: GCCACACTATCATAACCACTA	94°C for 3 min	94°C for 30 s	55°C for 30 s	72°C for 30 s	72°C for 4 min	[25]
icaA	1315	P1: CCTAACTAACGA AAGGTAG P2: AAGATATAGCGATAAGTG C	94°C for 3 min	94°C for 1.5 min	56.5°C for 45 s	72°C for 3 min	72°C for 3 min	[26]
icaD	381	P1: AAA CGT AAG AGA GGT GG P2: GGC AAT ATG ATC AAG ATA C	94°C for 3 min	94°C for 1.5 min	49°C for 30 s	72°C for 3 min	72°C for 3 min	[26]

(30 µg), oxacillin (1 µg), cefoxitin (30 µg), penicillin (10 µg) and fusidic acid (10 µg). The resistance incidence pattern was then analysed for the presence of biofilm-producing genes.

Data analysis

Data analysis was majorly carried out using percentages (%) of descriptive statistics, while the discrepancies percentage of phenotypic and genotypic identification was performed using the formula reported by Rasheed et al. [17].

$$\text{Phenotypic discrepancies } \% = \frac{\text{no. of isolates found negative for resistance gene from disc diffusion resistant isolates}}{\text{total number of resistant isolates found from disc diffusion test} \times 100}.$$

$$\text{Genotypic discrepancies } \% = \frac{\text{no. of isolates found positive for resistance gene from disc diffusion sensitive isolates}}{\text{total number of sensitive isolates found from disc diffusion test} \times 100}.$$

RESULTS

Prevalence of subclinical mastitis and *S. aureus*

The prevalence of SCM through CMT was recorded to be 59.11% (227/384) from apparently healthy bovines in three districts. The highest prevalence of SCM was noted to be 75.78% (97/128) in district Muzaffargarh followed by 61.72% (79/128) in district Faisalabad, and 39.84% (51/128) in district Rawalpindi. Upon the bacteriological, biochemical and molecular testing, an overall prevalence of *S. aureus* was noted to be 46.69% (106/227). The highest prevalence of 50.52% was found in district Muzaffargarh, followed by district Faisalabad (45.57%) while the lowest prevalence of 41.18% was recorded in district Rawalpindi.

Prevalence of MRSA and MSSA

The prevalence of MRSA was noted to be 44.33% (47/106) as these isolates presented resistance to the cefoxitin disc on phenotypic methods while the remaining sensitive isolates 55.66% (59/106) were categorised as phenotypic MSSA. Molecular investigation of the phenotypic MRSA revealed that 46.80% (22/47) of the isolates were harbouring the *mecA* gene while 53.19% (25/47) of the isolates were negative for the gene. Similarly, 33.89% (20/59) of the phenotypic MSSA was also carrying the *mecA* gene while 66.10% (39/59) of the isolates were negative for the resistant gene (Table 2).

Prevalence of BRSA and BSSA

Considering the β-lactam resistance pattern, 58.49% (62/106) of the isolates showed resistance to penicillin disc while 41.50% (44/106) isolates were sensitive to it and were termed as phenotypic BRSA and BSSA, respectively. *blaZ* gene analysis showed that 37.09% (23/62) of phenotypic BRSA isolates were positive for the resistance gene, while 62.91% (39/62) were not carrying it. In the same way, 36.36% (16/44) of the phenotypic BSSA isolates were also harbouring resistance genes, while 63.64% (28/44) were found negative towards that gene (Table 2).

Prevalence of VRSA and VSSA

For vancomycin resistance, 20.75% (20/106) isolates presented resistance to vancomycin disc and were categorised as phenotypic VRSA, while 79.24% (82/102) showed sensitivity to disc diffusion and were termed as

TABLE 2 Phenotypic and genotypic incidence of resistant strains.

Resistance gene	Class	Phenotypic (%)	Genotypic (%)	
			P[a]	*N*[b]
mecA	MRSA	47 (44.33)	22 (46.80)	25 (53.19)
	MSSA	59 (55.66)	20 (33.89)	39 (66.10)
blaZ	BRSA	62 (58.49)	23 (37.09)	39 (62.91)
	BSSA	44 (41.50)	16 (36.36)	28 (63.64)
vanB	VRSA	22 (20.75)	8 (36.36)	14 (63.63)
	VSSA	84 (79.24)	16 (19.04)	68 (80.95)
tetK	TRSA	38 (35.84)	10 (26.31)	28 (73.69)
	TSSA	68 (64.16)	13 (19.11)	55 (80.89)
accA-aphD	ARSA	32 (30.18)	06 (18.75)	26 (81.25)
	ASSA	74 (69.82)	08 (10.81)	66 (89.19)

Abbreviations: ARSA, aminoglycoside-resistant *Staphylococcus aureus*; ASSA, aminoglycoside-sensitive *S. aureus*; BRSA, β-lactam-resistant *S. aureus*; BSSA, β-lactam-sensitive *S. aureus*; MRSA, methicillin-resistant *S. aureus*; MSSA, methicillin-sensitive *S. aureus*; TRSA, tetracycline-resistant *S. aureus*; TSSA, tetracycline-sensitive *S. aureus*; VRSA, vancomycin-resistant *S. aureus*; VSSA, vancomycin-sensitive *S. aureus*.

[a]Resistant or sensitive isolates harbouring resistance gene.

[b]Resistant or sensitive isolates without resistance gene.

phenotypic VSSA. Genotypic investigation showed 36.36% (8/22) of phenotypic VRSA isolates bearing the *vanB* gene, while 63.63% (14/22) presented no resistance gene. In the same way, 19.04% (16/84) of phenotypic VSSA were also positive for the *vanB* gene, while 80.95% (68/84) were not carrying the resistance gene (Table 2).

Prevalence of TRSA and TSSA

Phenotypic resistance to tetracycline antibiotic disc was shown by 35.84% (38/106) of the isolates, while 64.16% (68/106) were recorded as sensitive and were classified as phenotypic TRSA and TSSA, respectively. Gene-level investigation showed that 26.31% (10/38) of the isolates were genotypically positive for the *tetK* gene while 73.69% (28/38) were not carrying the gene. Similarly, 19.11% (13/68) isolates were also carrying the resistance gene, while 80.89% (55/68) isolates did not depict the resistance gene (Table 2).

Prevalence of ARSA and ASSA

The prevalence of ARSA was found to be 30.18%, as (32/106) of the isolates showed resistance to gentamicin antibiotic disc and were classified as phenotypic ARSA while 69.82% (74/106) were termed as phenotypic VSSA due to sensitivity to gentamicin disc. The *accA-aphD* gene was present in 18.75% (06/32) phenotypic-resistant isolates, while 81.25% (26/32) isolates were genotypically negative for the gene. Similarly, 10.81% (08/74) of the phenotypically sensitive isolates were genotypically positive, in the meantime, 89.19 (66/74) were found to be genotypically absent for resistance (Table 2).

Prevalence of discrepancies

The maximum phenotypic discrepancy was recorded for the *acca-aphD* gene as 81.25% followed by 73.69%, 63.63%, 62.91% and 53.19% for *tetK*, *vanB*, *blaZ* and *mecA*, respectively. In the meantime, the highest genotypic discrepancy was shown by the *blaZ* gene at 36.36%, *mecA* gene at 33.89%, *tetK* at 19.11%, *vanB* at 19.04% and *accA-aphD* at 10.81% (Table 3).

Genotypic prevalence of biofilm-producing *S. aureus*

Gene-level investigation through PCR revealed that 49.06% (52/106) isolates were carrying the biofilm-producing *icaA* gene, while 40.57% (43/106) contained the *icaD* gene.

TABLE 3 Discrepancies between identification of multidrug resistance genes harbouring *Staphylococcus aureus*.

Targeted gene	Discrepancies % Phenotypic	Discrepancies % Genotypic	Ratio[a]
mecA	53.19	33.89	1.57
blaZ	62.91	36.36	1.73
vanB	63.63	19.04	3.34
tetK	73.69	19.11	3.86
accA-aphD	81.25	10.81	7.52

[a]Ration = phenotypic discrepancies/genotypic discrepancies.

Intercellular adhesion genes associated with resistance expression

Genotypic analysis of the intercellular adhesion genes (*icaA* and *icaD*) revealed that 68.09% (32/47) of the phenotypic MRSA isolates were harbouring either of the genes, while the remaining 31.91% (15/47) were negative for both the genes. Similarly, for phenotypic BRSA, 45 out of 62 (72.58%) were carriers of either *icaA* or *icaD*, while 27.42% (17/62) carried none of these. In the case of phenotypic VRSA 14/22 (63.64%) isolates were found positive for either gene while 36.36% of isolates (8/22) were negative for both genes. For phenotypic TRSA isolates, 21/38 (55.26%) were harbouring these genes, while 17/38 (44.74%) were negative for them. Out of 32 isolates, 19 (59.37%) samples showed a gene in their genetic material, while the remaining 40.63% (13/32) isolates were found negative towards that gene.

Intercellular adhesion genes-associated resistance ratio showed that the presence of any of these genes was expressing an enhanced rate of phenotypic resistance in the *S. aureus* isolates. In the case of phenotypic MRSA, there is 2.13 times increased expression of resistance against methicillin among isolates harbouring any of these intercellular adhesion genes contrary to the isolates with none of these genes. Similarly, 2.65, 1.75, 1.24 and 1.46 times compounding resistance was observed in phenotypic BRSA, VRSA, TRSA and ARSA isolates, respectively, when these isolates were carrying any of these genes as compared to when they were found negative for them (Table 4).

Phenotypic resistance pattern of MDRSA

An antibiogram of isolates either positive or negative for both or any one of the biofilm-producing genes was obtained, and it reported a minimum percentage of isolates, that is, 2.22% showed resistance against six different antibiotics, including oxacillin, cefoxitin, tylosin, ciprofloxacin, tetracycline and levofloxacin and these isolates were positive for both *icaA* and *icaD* genes. While 15.56% of isolates that were positive for *icaD* and negative for the *icaA* gene represented resistance against gentamicin, amikacin, tetracycline and amoxicillin (Table 5). Among the tested isolates against

TABLE 4 Role of intercellular adhesion genes in expression of antibiotic resistance.

Phenotypic (*n* = no. of isolates)	Interceullar adhesion gene[a]		Intercelluar adhesion genes associated resistance ratio[b]
	Present	Absent	
MRSA (*n* = 47)	32	15	2.13
BRSA (*n* = 62)	45	17	2.65
VRSA (*n* = 22)	14	8	1.75
TRSA (*n* = 38)	21	17	1.24
ARSA (*n* = 32)	19	13	1.46

Abbreviations: ARSA, aminoglycoside-resistant *Staphylococcus aureus*; BRSA, β-lactam-resistant *S. aureus*; MRSA, methicillin-resistant *S. aureus*; TRSA, tetracycline-resistant *S. aureus*; VRSA, vancomycin-resistant *S. aureus*.
[a] *icaA* or *icaD*.
[b] Intracellular adhesion genes associated resistance ratio = number of resistant isolates with intercellular adhesion gene/number of resistant isolates without intercellular adhesion gene.

TABLE 5 Phenotypic resistance pattern of biofilm-producing genes harbouring *Staphylococcus aureus*.

Interceluar adhesion genes			
icaA	*icaD*	Resistance to antibiotics	Incidence (%)
+	+	OX + FOX + TY + CIP + T + LEV	5 (11.11)
+	+	FOX + CN + AK + T + MXF	4 (8.89)
−	+	CN + AK + T + AXE	7 (15.56)
+	+	OX + LNZ + CN + MXF	3 (6.67)
+	+	FOX + OX + CN + T + AK	2 (4.44)
−	+	OX + CIP + T + MXF	1 (2.22)
+	−	OX + FOX + CN + AK	3 (6.67)
+	−	T + CN + AK + TY	2 (4.44)
−	+	FOX + CN + T + AXE	1 (2.22)
−	+	OX + FOX + AXE + LNZ	1 (2.22)
+	−	FOX + OX + LNZ + AK + LEV + MXF	5 (11.11)
+	−	FA + MXF + LNZ	3 (6.67)
−	−	T + TY + LEV	2 (4.44)
−	−	OX + CN + AK	4 (8.89)
−	−	OX + FOX+TY	2 (4.44)

Abbreviations: AK, amikacin; AXE, amoxicillin; CIP, ciprofloxacin; CN, gentamicin; FA, fusidic acid; FOX, cefoxitin; LEV, levofloxacin; LNZ, linezolid; MXF, moxifloxacin; Ox, oxacillin; P, penicillin; T, oxytetracycline; TY, tylosin.

various antibiotics, 42.45% (45/106) of *S. aureus* strains were considered multiple drug resistant (MDR), while 57.55% of isolates did not show MDR pattern on a phenotypic basis. The isolates were declared MDR as per the findings of Javed et al. [28].

DISCUSSION

S. aureus is the most common and rapidly evolving pathogen having bidirectional diffusion between humans and animals [29, 30]. The resistance expression is modulated by the presence of several resistance-encoding genes and the formation of physical barriers around the pathogen in the form of biofilm [31, 32]. This study aimed to investigate the phenotypic and genotypic antimicrobial-resistant pathogen along with an assessment of intercellular adhesion genes' impact on the expression of resistance in *S. aureus* isolates.

The current study revealed that *S. aureus* was associated with 46.69% (106/227) of the SCM infection in the sampled bovine population. This high level of association can be explained based on the poor hygiene during the husbandry and milking processes at the farm, as well as *S. aureus* being the commensal organism that can transfer from farm workers to the animals while maintaining improper working conditions. The prevalence of *S. aureus* isolated from bovines has been reported previously in Pakistan at 56.12% [33], 48.82% [14], 41.41% [32], 37.14% [19], 34.2% [34], 28.70% [35] and 17.96% [36] as per different studies. One study reported an even higher prevalence of 71.52% in Pakistan [37]. The mean variation in the *S. aureus* association with SCM can be due to variable sample size and the varied management practices at an urban or rural rearing centre along the climatic variations.

Phenotypic assessment of the antibiotic resistance showed that penicillin was found to be dealing with the most resistance, as 48.49% of the isolates were found to be resistant to it, followed by cefoxitin at 44.33%, tetracycline at 35.84%, gentamicin at 30.18% and vancomycin as 20.75%. The highest incidence of penicillin resistance is also reported by several other studies [38–40]. A similar order of phenotypic antibiotic resistance was observed by Mekuria et al. [41] and was correlated with increased levels of use, overuse and misuse of antibiotics in farming either as treatment or prophylactic approach. Vancomycin is not directly used in the livestock sector in Pakistan, but the incidence of vancomycin resistance can be due to the diffusion of resistance-modulating factors from the workers to the animals [42].

Genotypic antibiotic resistance evaluation sheds light on the presence or absence of several antibiotic resistance-modulating genes. These genes alone and in combination with several other factors influence the expression of phenotypic resistance either by the production of enzymes that destroy the antibiotics or by expressing an altered form of structure on which the antibiotic acts. The genotypic investigation revealed that *mecA* was the most prevalent resistance gene at 46.80%, followed by *blaZ* at 37.09%, *vanB* at 36.36%, *tetK* at 26.31% and *accA-aphD* at 18.75%, which accounts for MRSA, BRSA, TRSA and ARSA isolates, respectively. These genotypic prevalence results are not in line with the phenotypic incidence of the genes and the variation can be due to discrepancies between methods of identification or other resistance-modulating factors. The highest level of *mecA* gene presence can be explained by the overlapping mechanism of action displayed by *mecA* and *blaZ* genes for the expression of resistance. A somewhat similar order of gene harbouring was also reported by Pajohesh et al. [38], but he reported the highest percentage of *blaZ* gene, followed by others, which is different from our study. The prevalence of MRSA having *mecA* gene isolated from different livestock species has been documented in the previous studies conducted in goats [28], bovines [14], sheep [40] and equines [43] in different regions of Pakistan, which are in the range of the current study findings of MRSA. The current study's prevalence of VRSA isolated from bovine origin is higher than in previous studies [19, 42]. Similarly, according to the research conducted by Bakht et al. [36], *S. aureus* isolates of bovine reproductive tract origin depicted various ranges of antibiotic-resistant strains like 18.84% MRSA, 27.54% TRSA and 36.23% ARSA, respectively, which differs from the current study findings.

Biofilm production by *S. aureus* is one of its mechanisms for evasion from the immune response of the host and leads to persistent infection along with the spread of resistance genes to other bacteria by reducing the penetrative abilities of the antibiotics [44–46]. Intercellular adhesion (*ica*) gene locus is associated with biofilm formation with almost equal levels of *icaA* and *icaD* gene [26, 47]. The current study reported the level of the *icaA* gene at 49.06% and *icaD* gene at 40.57% of the isolates which is in correlation with the results reported by other studies [46, 48–50] but lower than the previous study [51]. Our results deviated from the studies of [47] and [52] which reported that *icaD* is more prevalent than the *icaA* gene. This variation in the gene prevalence can be explained based on mutation or change in DNA sequence that can hinder the amplification of genes in a few bacterial isolates giving false negative results for any of the genes [47].

The phenotypic assessment of antibiotic resistance expression revealed that the presence of any of the intercellular adhesion genes, *icaA* or *icaD*, enhanced the probability of the expression of the resistance by that isolate. Results supporting the increased antibiotic resistance by genotypic biofilm-producing isolates are reported by Ahmed et al. [32] and Manandhar et al. [53]. The enhanced probability of resistance expression by phenotypic-resistant isolates in the presence of biofilm-producing genes can be explained by additional physical resistance incorporated by the biofilm covering of the bacterium as well as the provision of adhesion to the host surface.

This study also highlighted the concept of discrepancies, which are responsible for the difference in prevalence of these resistant isolates both phenotypically and genotypically. *S. aureus* harbouring resistance genes is identified using phenotypic and genotypic techniques; however, the outcomes of these approaches are not always concordant. In this study, phenotypic discrepancies, which are the number of resistant isolates presented by the disc diffusion method that returned negative for the resistant gene, were found, which indicates that there are multiple resistance determinants [37, 54, 55]. These discrepant results are also referred to as adaptive immunity [56]. These contradictory findings suggest that a single gene's presence or absence does not accurately identify isolates that are sensitive or resistant. These findings also imply that resistance genes are not the exclusive cause of antibiotic resistance and that there are several processes underlying antibiotic resistance.

In this study, genotypic discrepancies, which is the number of sensitive isolates presented by the disc diffusion method that returned positive for the resistant gene, were also found and it demonstrated that certain gene regulators or repressors aid in the expression of the gene; hence, these inconsistent findings are explained by mutations or alterations in gene regulators [57, 58]. These findings are in line with a study by Emaneini et al. [59], which identified four isolates that were tetracycline-sensitive but positive for *tetK* genes. These findings imply that the phenotypic expression of resistance is not solely dependent on the presence of a resistance gene but also requires the presence of certain regulatory factors. The *tetK* gene may be silenced in phenotypic expression, according to a study [60]. This study also brought attention to the phenomenon of antibiotic resistance determinants being silenced. The higher discrepancies ratio or the percentage change of phenotypic characteristics over genotypic characteristics indicates that several other mechanisms or genes are responsible for causing the antibiotic resistance in the organism [17].

CONCLUSION

The current study reported a high incidence of antibiotic resistance along with biofilm-encoding genes in the local *S. aureus* isolates of bovine origin. The high relevance of biofilm production with the expression of antibiotic resistance was also evident in our study. The presence of biofilm genes exerted an enhanced rate of antibiotic resistance expression. The current study also highlighted significant discrepancies between the phenotypic and genotypic resistance identification methods.

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CONFLICT OF INTEREST STATEMENT

The authors declared that there are no conflicts of interest.

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